

NCMC Final Paper: On the Hausdorff Dimension and Box-Counting Methods

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Abstract

Your abstract.

1 Introduction

1.1 Motivation

Classical geometry provides simple notions of dimension for familiar objects such as line segments, squares, and cubes. However, many naturally occurring objects exhibit highly irregular structure and cannot be adequately described by these traditional ideas. Fractal geometry was developed to study such objects, and the Hausdorff dimension has become one of its most important tools for measuring geometric complexity.

Although the Hausdorff dimension provides a rigorous mathematical definition of dimension, its construction relies on ideas from measure theory and can be challenging for those encountering the subject for the first time. At the same time, practical applications often require computational methods for estimating the dimensions of complicated sets. The box-counting method offers a simple and effective approach that can be applied to both mathematical fractals and real-world data.

The purpose of this paper is to introduce the mathematical foundations of the Hausdorff dimension in an accessible manner. We develop the necessary concepts from measure theory, establish several fundamental properties of the Hausdorff dimension, and compute the dimensions of several classical self-similar sets. We then introduce the box-counting method and apply it to estimate the dimensions of irregular coastlines, illustrating how theoretical ideas from fractal geometry can be used to analyze naturally occurring structures.

1.2 Intuition

We provide some examples to build our intuition for the Hausdorff dimension. Consider splitting a segment of length 1 into two equal pieces of length $\frac{1}{2}$. The smaller line is half the length of the original line. A square of side-length 1 can be split into four smaller squares of side-length $\frac{1}{2}$. A cube of side-length 1 can be split into eight smaller cubes of side-length $\frac{1}{2}$. Similarly, a Sierpinski triangle of length 1 is made of three copies of itself with length $\frac{1}{2}$.

All of these geometric objects are characterized by a certain measure. The smaller line is half the *length* of the original line, the smaller square is a quarter of the area of the original square, and the smaller cube is one eighth of the volume of the original cube. The Hausdorff measure generalizes this concept to describe the measure of fractals like the Sierpinski triangle.

More intuitively, let us define the mass as a concept of measure. For example, when we scale a length by $\frac{1}{2}$, the mass is also scaled by $\frac{1}{2}$.

We claim that the Sierpinski triangle has a mass scalar of $\frac{1}{3}$ with a length scalar of $\frac{1}{2}$. Note that it takes three smaller triangles to form the original. Also note that the mass scalar is an integer power of the length scalar for the segment, square, and cube. In fact, we claim that the exponent is the

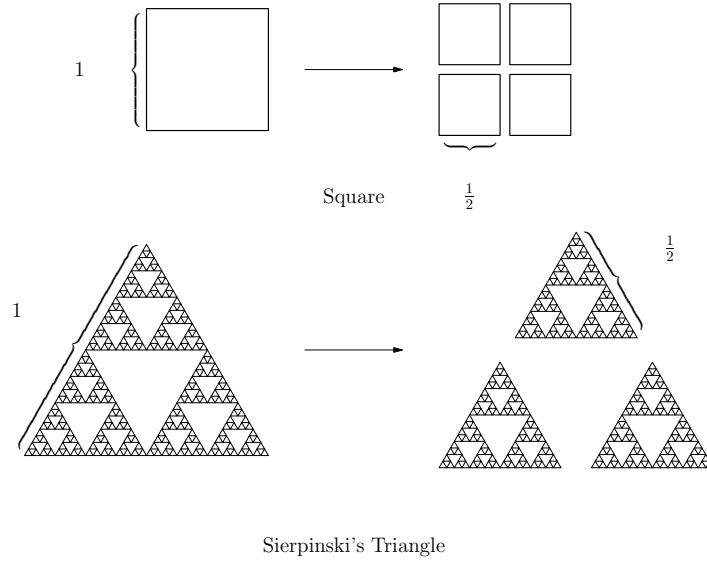


Figure 1: The geometric objects of a square (top) and a Sierpinski Triangle (bottom) are split into scaled shapes with $\frac{1}{2}$ length. The change in mass differs.

Object	Length Scalar	Mass Scalar	Dimension
Segment	$\frac{1}{2}$	$\frac{1}{2}$	1
Square $\frac{1}{2}$	$\frac{1}{4}$	2	
Cube	$\frac{1}{2}$	$\frac{1}{8}$	3
Sierpinski Triangle	$\frac{1}{2}$	$\frac{1}{3}$	$\log_2 3$

Table 1: Length and mass scaling factors under self-similar subdivision by $\frac{1}{2}$. We claim the dimension of the Sierpinski Triangle is $\log_2 3$.

dimension of each shape. The length is 1-dimensional, and $(\frac{1}{2})^1 = \frac{1}{2}$. The square is 2-dimensional, and $(\frac{1}{2})^2 = \frac{1}{4}$. Similarly, an object being 3-dimensional implies that when the length scalar is $\frac{1}{2}$, then the mass scalar is $(\frac{1}{2})^3 = \frac{1}{8}$. Since the Sierpinski triangle is self-similar, we want its mass to scale $\frac{1}{3}$ with a length of $\frac{1}{2}$. We solve for the dimension d with the equation

$$\left(\frac{1}{2}\right)^d = \frac{1}{3}$$

and we have $d = \log_2 3 \approx 1.585$. The measure of the Sierpinski triangle is approximately the 1.585-dimensional analog of length. In general, the formula

$$d = \frac{\log(\text{mass scalar})}{\log(\text{length scalar})}$$

is a direct consequence of Moran's equation for self-similar sets (CITE HERE!).

2 Hausdorff Dimension

We begin by introducing the fundamental concepts of measure theory that form the basis for the construction of Hausdorff measure. These concepts provide a framework for extending traditional notions of length, area, and volume to irregular fractal sets.

2.1 Outer Measures

Definition 1 (Outer Measure). Let X be a set, and let $\mathcal{P}(X)$ denote the collection of all subsets of X . A map

$$\mu^* : \mathcal{P}(X) \rightarrow [0, +\infty]$$

is called an **outer measure** on X if it assigns a "size" to every subset of X satisfying the following three natural conditions:

- **The empty set has size zero:**

$$\mu^*(\emptyset) = 0.$$

- **Monotonicity**, a subset cannot be measured as larger than a set containing it:

$$A \subseteq B \implies \mu^*(A) \leq \mu^*(B).$$

- **Countable subadditivity**, covering a union of sets can never provide a lower size. For any sequence $(A_n)_{n \in \mathbb{N}}$ of subsets of X ,

$$\mu^*\left(\bigcup_n A_n\right) \leq \sum_n \mu^*(A_n).$$

Unlike an ordinary measure, an outer measure does not require the sizes of disjoint sets to add exactly. Instead, it only requires that the size of their union is at most the sum of their individual sizes. This weaker condition makes it possible to define an outer measure for every subset of X , even sets with very irregular structure.

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The three axioms above capture the basic properties that any reasonable notion of size should satisfy. The empty set should have size zero, enlarging a set should not decrease its size, and covering a set by simpler pieces should not underestimate its total size. These properties are general enough to apply even to highly irregular sets, making outer measures a natural starting point for constructing more refined notions of measure.

Proposition 1. Let μ^* be an outer measure. Then for any subsets $A, B \subseteq X$,

$$\mu^*(A \cup B) \leq \mu^*(A) + \mu^*(B).$$

Proof. This is the special case of countable subadditivity obtained by taking

$$A_1 = A, \quad A_2 = B, \quad A_n = \emptyset \quad (n \geq 3).$$

Since $\mu^*(\emptyset) = 0$,

$$\mu^*(A \cup B) = \mu^*\left(\bigcup_{n=1}^{\infty} A_n\right) \leq \sum_{n=1}^{\infty} \mu^*(A_n) = \mu^*(A) + \mu^*(B).$$

□

2.2 Gauge Functions

Consider the rectangle with length ℓ and width w . If we measure this shape with length ($s = 1$), there are infinitely many vertical and horizontal line segments in the square. If we measure this shape with area ($s = 2$), we find its area as $0 < \ell w < \infty$. If we measure this shape with volume ($s = 3$), we find 0, since an area contains no volume.

Definition 2 (Gauge Function). A gauge function is a function

$$\varphi : [0, \infty) \rightarrow [0, \infty)$$

such that:

1. $\varphi(0) = 0$
2. φ is non-decreasing
3. $\varphi((0, \infty)) \subseteq (0, \infty)$
4. $0 = \varphi(0) = \lim_{t \rightarrow 0} \varphi(t)$ (φ continuous at 0)

For example, $\varphi(t) = t^s$ is a gauge function for every $s > 0$. This is the most important family of gauge functions because it gives rise to the s -dimensional Hausdorff measure. When covering a set by subsets $\{U_i\}_{i=1}^\infty$, each set contributes a cost of

$$\varphi(\text{diam}(U_i)) = (\text{diam}(U_i))^s.$$

The total cost of a cover is then the sum of these values, and the Hausdorff measure is obtained by taking the infimum over all sufficiently fine covers. Varying the exponent s changes how heavily large and small sets are weighted, allowing the measure to detect the dimension of fractal sets.

2.3 Hausdorff Measure

Definition 3. Let (X, d) be a metric space and $\varphi : (0, \infty) \rightarrow (0, \infty)$ be a gauge function. Let $\delta > 0$ and $A \subseteq X$. A countable cover $\{C_i\}_{i=1}^\infty$ of A such that

$$\sup\{\text{diam}(C_i) : i \in \mathbb{N}\} \leq \delta$$

is called a δ -cover of A . Define then

$$H_\varphi^\delta(A) := \inf \left\{ \sum_{i=1}^\infty \varphi(\text{diam}(C_i)) \mid \{C_i\}_{i=1}^\infty \text{ a } \delta\text{-cover of } A \right\}$$

Remark 1 (Explanation of the definition). This definition builds up the notion of a δ -cover in several steps.

- **The metric space (X, d) .** This is just a set X together with a notion of distance $d(x, y)$ between any two points $x, y \in X$. A common example is $X = \mathbb{R}^n$ with the usual Euclidean distance, but X could be far more general.
- **The gauge function φ .** A gauge function takes a positive number (thought of as a “size”) and returns another positive number (a “weighted size”). In the classical construction of Hausdorff measure, one uses $\varphi(t) = t^s$ for some fixed exponent $s \geq 0$; allowing a general φ lets us measure sets more finely than any single power t^s can, which is why this construction is often called a Hausdorff measure with respect to a gauge function (or a generalized Hausdorff measure).
- **The diameter of a set.** For a set $C_i \subseteq X$, $\text{diam}(C_i) := \sup\{d(x, y) : x, y \in C_i\}$ is the largest distance between two points of C_i . It measures how “spread out” C_i is.
- **A countable cover.** A collection $\{C_i\}_{i=1}^\infty$ of subsets of X is a cover of A if $A \subseteq \bigcup_{i=1}^\infty C_i$, i.e. every point of A lies in at least one C_i . There can be finitely or infinitely many sets C_i , and they are allowed to overlap.
- **The δ -restriction.** The condition $\sup\{\text{diam}(C_i) : i \in \mathbb{N}\} \leq \delta$ says that every piece C_i in the cover has diameter at most δ . In other words, we are only allowed to cover A using “small” pieces, none of which is bigger (in diameter) than δ . Such a cover is called a δ -cover.
- **The quantity $H_\varphi^\delta(A)$.** For each admissible δ -cover $\{C_i\}$, we compute the total weighted size $\sum_{i=1}^\infty \varphi(\text{diam}(C_i))$, i.e. we apply the gauge function to the diameter of each piece and add up the results. Then $H_\varphi^\delta(A)$ is defined as the infimum of this sum over all possible δ -covers of A : it is the smallest total weighted size achievable when covering A using pieces of diameter at most δ .

Intuitively, $H_\varphi^\delta(A)$ is our best attempt at measuring the “ φ -size” of A , using a bound of size δ . We are forbidden from using pieces coarser, or with a larger diameter, than δ , and among all fine-enough covers we take the most efficient (smallest-total) one. Shrinking δ only removes coarser covers from consideration, so $H_\varphi^\delta(A)$ can only increase (or stay the same) as δ decreases. This monotonicity is what allows one to later define the Hausdorff measure. In fact, for $\varphi(t) = t^s$, $H_\varphi(\cdot) = H_s(\cdot)$ is called the s -dimensional Hausdorff measure.

Several sources also provide a Hausdorff measure for generic sets with an implied gauge function. Let $F \subseteq \mathbb{R}^n$, $s > 0$ with $s \in \mathbb{R}$. For any $\delta > 0$ we define

$$\mathcal{H}_\delta^s(A) = \inf \left\{ \sum_{i=1}^{\infty} |U_i|^s \mid U_i \text{ is a } \delta\text{-cover of } F \right\}$$

Note that this alternative definition without the gauge function allows us to write the s -dimensional Hausdorff measure as a limit:

$$\mathcal{H}^s(\cdot) = \lim_{\delta \rightarrow 0^+} \mathcal{H}_\delta^s(\cdot)$$

which is equivalent to the previous definition.

2.4 Hausdorff Dimension

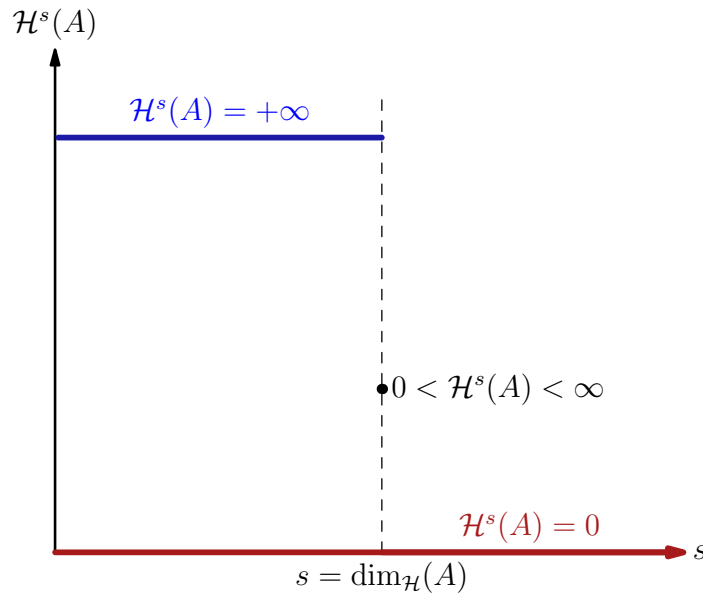


Figure 2: The Hausdorff dimension is the critical exponent at which the s -dimensional Hausdorff measure transitions from infinity to zero.

Definition 4 (Hausdorff Dimension). *The Hausdorff dimension of a set A can be defined by*

$$\dim_{\mathcal{H}}(A) = \inf\{s \geq 0 : \mathcal{H}^s(A) = 0\} = \sup\{s \geq 0 : \mathcal{H}^s(A) = +\infty\}.$$

The behavior of the Hausdorff dimension is visually depicted in Figure 2.

3 Properties and Applications

3.1 Moran's Theorem

Moran's Theorem, originally proved by P. A. P. Moran, allows us to find the Hausdorff dimension of self-similar sets more easily.

Theorem 1 (Moran's Theorem). *Let $S_1, \dots, S_N$ be similarity contractions on $\mathbb{R}^n$ with contraction ratios*

$$0 < r_1, r_2, \dots, r_N < 1.$$

Suppose the iterated function system satisfies the Open Set Condition (OSC). Let A be the unique nonempty compact set satisfying

$$A = \bigcup_{i=1}^N S_i(A).$$

Then the Hausdorff dimension of A is the unique real number s satisfying

$$\sum_{i=1}^N r_i^s = 1.$$

Moreover,

$$0 < \mathcal{H}^s(A) < \infty,$$

where $\mathcal{H}^s$ denotes the s -dimensional Hausdorff measure.

The proof of Moran's Theorem is out of the scope of this paper and can be found in the above source.

Note that when all copies share the same contraction ratio $r_1 = r_2 = \dots = r_N = r$, the sum collapses to N identical terms, giving the simplified (uniform) form of Moran's equation:

$$\sum_{i=1}^N r_i^s = N r^s = 1$$

We solve for s to find

$$s = \frac{\log N}{\log(1/r)}$$

for a self-similar set made of N identical-ratio copies scaled by r . This formula immediately recovers known results. The Sierpinski triangle has $N = 3$ self-similar copies with a contraction ratio of $r = \frac{1}{2}$. We calculate the Hausdorff dimension as $s = \frac{\log 3}{\log(\frac{1}{2})}$.

3.2 Cantor Set

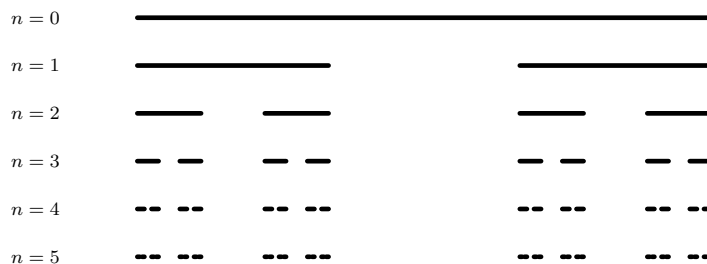


Figure 3: The Cantor Set is an example of a uniform self-similar set. It is constructed by an Iterated Function System (IFS).